

AMS 572 Project Report

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Due: December 5, 2025 10:00am

1 Introduction

Nearly two decades since its introduction in 2009, Bitcoin has gained mainstream traction as an investment asset. This digital currency is held by major heads funds[1], Fortune 500 companies[2], and national banks[3]. Additionally, Bitcoin has gained acceptance as an everyday currency with many retailers now accepting it as payment. Most incredibly, Bitcoin was even declared a legal-tender currency of El Salvador in September 2021[4]. As Bitcoin has become more accepted, retail investors calmore to buy and trade this new asset. Today, you can hold Bitcoin in digital wallets, investment and retirement accounts[5], as well as purchasing shares of Bitcoin-focused exchange-traded funds (ETFs)[6].

Moreover, the meteoric rise in price per bitcoin has spawned a new class of digital assets called cryptocurrencies. Aside from Bitcoin, these cryptocurrencies follow one of two paradigms. The first paradigm describes coins that focus on improving some aspects of the technology such as energy efficiency, transaction speed, or added utility. These coins utilize their own blockchain networking structure leading to their name as “native coins”. The second type of coin attempts to capitalize on the speculation and excitement around Bitcoin’s price by creating eye-catching imitations. These coins use existing blockchain networks and are often themed around a specific internet trend or personality to garner attention. In online circles, these types of coins have earned the title of “meme coin” as they make no novel improvements to the underlying structure.

Due to its mainstream adoption in traditional financial markets by both institutional and retail investors, does Bitcoin's price follow stock market movements? Is Bitcoin truly a new investment asset, or is its price so deeply entangled with traditional investments that it has lost its novelty? The answer to this question has serious ramifications for portfolio allocations. If Bitcoin's movements are unrelated to the stock market, it can be a valuable tool to diversify portfolios and reduce risk. However, if Bitcoin's price follows the movements of the stock market, its utility as an investment asset mirrors similar stocks in the digital sphere.

How do the price movements of Bitcoin proportionally affect Native or Meme coins? Are investors more excited about the underlying technology powering digital assets or are they seeking an avenue to wildly speculate? The answer to this question will help unravel the motivation fueling the cryptocurrency boom. This is because of the dual purpose of currency, being a store-of-value and a mode of facilitating transactions. Both of these purposes benefit from the price being predictable and stable. If the meme coins are more sensitive to changes in Bitcoin, it highlights their use as a speculative asset. Likewise, this result highlights the utility of native coins and investors bet on the underlying technological improvements.

1.1 Questions of Interest

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1.2 Description of the Data

The data for this analysis was imported from Yahoo Finance using the `quantmod` R package. This data contains the date, daily opening price, daily high price, daily closing price, and daily volume for each asset specified. The range of dates collected for this analysis ranges from November 1, 2023 to November 18, 2025. This data was then transformed to isolate the log-daily returns over this time period. Additionally, a dummy variable was added to categorize between the native coins and the meme coins. When combining the SPY and BTC data, non-trading days such as holidays and weekends were removed.

2 Background

In this section, we review some basic mathematical machinery with regard to financial time series.

2.1 Introduction to Financial Time Series

By a time series we mean a sequence of observations taken at regular time intervals. More formally, a **time series** is simply a sequence $\{x_t\}_{t \in T}$ of real numbers x_t , for $T = \{t_0, t_1, t_2, t_3, \dots, t_n\}$ some finite indexing set of time increments over a certain period¹. In the case of finance, our sequence $\{x_t\}_{t \in T}$ (which we may write equivalently as x_t for brevity, where the time index T is implicitly understood) represents the evolution of the price of a particular asset over time. To our notation

¹In the case of real-world data, this indexing set is finite. However, generally we can make T to be any finite, countably infinite, or even uncountably infinite subset of $\mathbb{R}$ (the real numbers).

consistent, we denote the price of a particular asset at the end of a trading day t by P_t .

When we study the evolution of an asset price, it is often useful to measure changes in price by computing the daily **rate of return** $\frac{P_t}{P_{t-1}}$ or, more preferably, the **daily log return** time series r_t of P_t , which is given by the formula

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right). \quad (1)$$

An example of a financial time series (i.e., the daily return of the S&P 500 stock market index), its daily return & daily log return time series is shown in Figure 1 below.

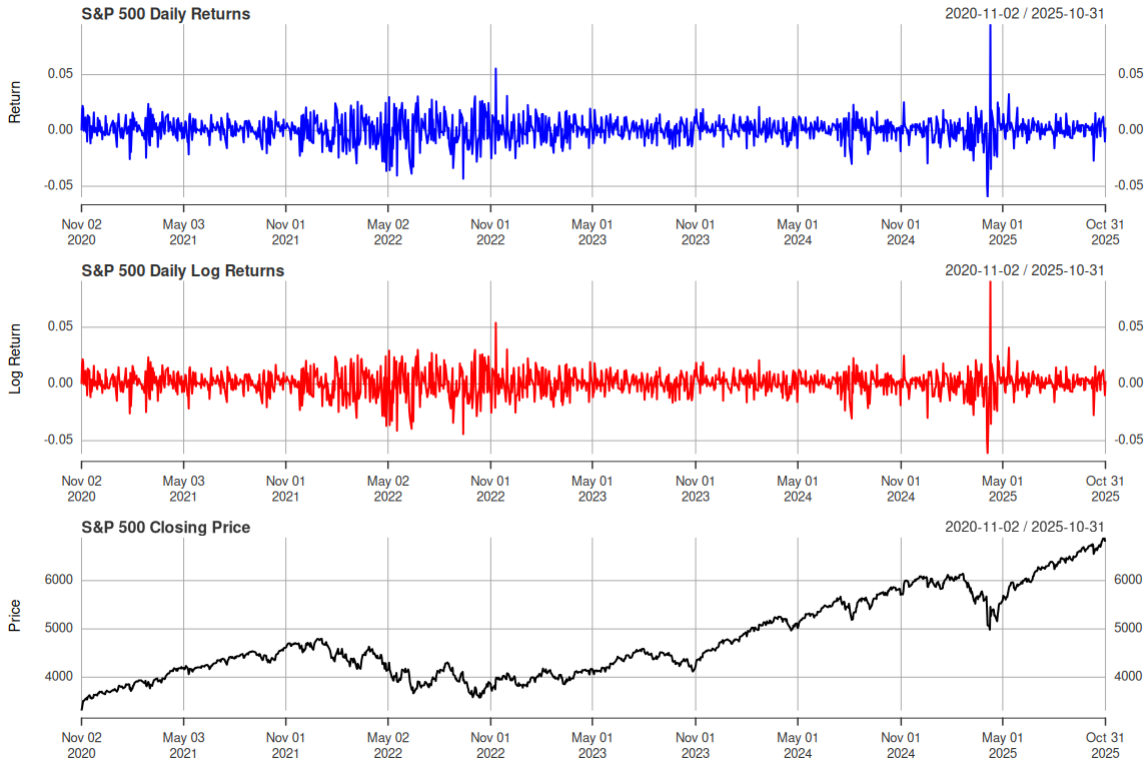


Figure 1: S&P 500 Daily Returns, Daily Log Returns, and Closing Prices between Nov 1, 2020 and Nov 1, 2025.

In what follows, we will use analyze the price of different cryptocurrencies and financial assets using log returns as in Equation 1.

2.2 Inference for the Correlation between Time Series

Recall that, if X and Y are continuous random variables with $E[X] = \mu_X$ and $E[Y] = \mu_Y$, variances $\text{Var}(X) = \sigma_X^2$ and $\text{Var}(Y) = \sigma_Y^2$, then their **covariance** is given by the formula

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

and subsequently their **correlation** is given by

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

How can we define the same notion of correlation for time series to asses a relationship between them? It turns out, if we are given two time series $\{x_t\}_{t=1}^N$ and $\{y_t\}_{t=1}^N$, we can use the **Pearson correlation coefficient**, defined in Equation 2 below as

$$\hat{\rho} = \frac{\sum_{t=1}^n (x_t - \bar{x})(y_t - \bar{y})}{\sqrt{\sum_{t=1}^n (x_t - \bar{x})^2} \sqrt{\sum_{t=1}^n (y_t - \bar{y})^2}}, \quad (2)$$

where $\bar{x}$ and $\bar{y}$ denote the sample means of $\{x_t\}_{t=1}^n$, $\{y_t\}_{t=1}^n$, respectively. I.e.,

$$\bar{x} = \frac{1}{n} \sum_{t=1}^n x_t, \quad \bar{y} = \frac{1}{n} \sum_{t=1}^n y_t.$$

When studying the relationship between two time series x_t and y_t , we can use the Pearson correlation coefficient in Equation 2 to construct a hypothesis test of $H_0 : \hat{\rho} = 0$ vs. $H_1 : \hat{\rho} \neq 0$ under the following assumptions:

1. The pairs (x_t, y_t) are independent across time: for $t_1 \in T$ and $t_2 \in T$, (x_{t_1}, y_{t_1}) and (x_{t_2}, y_{t_2}) are independent, and
2. (x_t, y_t) are jointly normally distributed.

Under these assumptions, we use the test statistic

$$T = \hat{\rho} \sqrt{\frac{n-2}{1-\hat{\rho}^2}}, \quad (3)$$

where T follows a t -distribution with $df = n - 2$ degrees of freedom under H_0 .

In practice, particularly when dealing with financial data, the assumptions 1 and 2 are often not well satisfied. For starters, the log return of any particular stock often *does not* follow a normal distribution, which violates assumption 2. Furthermore, the development of the price of a stock may not be generally independent across time. However, for purposes of simplification, it is often assumed to be the case in the theory financial modeling that stock prices are independently and identically normally distributed.

2.3 Linear Regression on Financial Time Series

For **Question 2**, we are interested in whether or not the price of memecoins track closer to the price of Bitcoin versus other, more serious nativecoins. We will study this question by performing a linear regression analysis. Let

$$\text{Memecoin}_i = \begin{cases} 1 & \text{if coin } i \text{ is a memecoin,} \\ 0 & \text{if coin } i \text{ is a nativecoin.} \end{cases}$$

We then fit the model

$$\text{Return}_{i,t} = \beta_0 + \beta_1 \text{BTC}_t + \beta_2 \text{Memecoin}_{i,t} + \beta_3 (\text{BTC}_t \times \text{Memecoin}_{i,t}) + \varepsilon_{i,t}. \quad (4)$$

We can interpret the coefficients in Equation 4 in the following way: β_1 can be viewed as the sensitivity of native coins to changes in the price of Bitcoin (notice that when $\text{Memecoin}_i = 0$ for a nativecoin i , the slope of Equation 4 is simply β_1). Moreover, β_3 can be viewed as the *difference* in sensitivity for memecoins versus nativecoins, since when $\text{Memecoin}_i = 1$ for a memecoin, the

slope is $\beta_1 + \beta_3$. With this in mind, if it turns out that $\beta_3 > 0$, we have that memecoins are more sensitive to the price of Bitcoin than nativecoins are.

In order to apply the linear regression analysis as in Equation 4, which has the general form $Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$, we require the following conditions to be satisfied by X, Z , and the interaction term XZ :

1. The conditional expectation of Y should be linear in the parameters, i.e., $\mathbb{E}[Y \mid X, Z] = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$,
2. the regressors X, Z , and XZ should be uncorrelated with the error term,
3. no regressor is a linear combination of the others,
4. $\text{Var}(\varepsilon \mid X, Z)$ is constant (this is called **homoskedasticity**),
5. errors are uncorrelated across observations (this means that there is **no autocorrelation**), and
6. ε is normally distributed.

Again, in the case of financial time series, assumptions 4 and 5 are often not satisfied. As a result, if we were to perform a standard linear regression analysis, we would not be able to trust our OLS standard errors and their associated p -values. To remedy this fact, we instead compute a different flavor of standard errors, called **heteroskedasticity and autocorrelation consistent standard errors**, which will provide us with valid inference results even when assumptions 4 and 5 are not satisfied.

To motivate and illustrate this alternative approach, consider the standard linear model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

for $\mathbf{Y} \in \mathbb{R}^n$ a vector of responses, $\mathbf{X} \in \mathbb{R}^{n \times p}$ a design matrix, where the i th row is denoted X_i , $\boldsymbol{\beta} \in \mathbb{R}^p$ is the vector of parameters, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the error vector, where under classical OLS

assumptions we assume $\text{Var}(\varepsilon_i) = \sigma^2$, where σ is a **constant**, and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. We already know that the standard OLS estimator of $\boldsymbol{\beta}$ is given by

$$\widehat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{Y}$$

which is unbiased provided $[E](\boldsymbol{\varepsilon}) = \mathbf{0}$ and the errors are independent of $\mathbf{X}$. If we want to perform the hypothesis test

$$H_0 : \beta_j = 0,$$

and compute a p -value, we compute the test statistic

$$t_j = \frac{\widehat{\beta}_j}{\text{SE}(\widehat{\beta}_j)} = \frac{\widehat{\beta}_j}{\sqrt{\widehat{\text{Var}}(\widehat{\boldsymbol{\beta}})}} \quad (5)$$

where $\widehat{\text{Var}}(\widehat{\boldsymbol{\beta}}) = \widehat{\sigma}^2 (X^T X)^{-1}$ is the sample variance of the OLS estimator. However, the problem is, this formula for the sample variance is *only* valid when the assumption that the errors are IID holds. In the case where this does not hold (in particular, in the case where the errors are correlated and heteroskedastic), we need to use a different estimator of the covariance matrix of $\widehat{\boldsymbol{\beta}}$, called the *Newey-West estimator*.

The Newey-West estimator of the covariance matrix of $\widehat{\boldsymbol{\beta}}$ is given by

$$\widehat{\text{Var}}_{\text{NW}}(\widehat{\boldsymbol{\beta}}) = (X^T X)^{-1} \left(X^T \widehat{\boldsymbol{\Omega}}_{\text{NW}} X \right) (X^T X)^{-1}, \quad (6)$$

where

2.4 Autocorrelation, Heteroskedasticity, and Robust Inference

3 Question 1

4 Question 2

5 Conclusion